

Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 7

The integral in this exercise converges. Evaluate the integral without using a table.

$$\int_{-\infty}^0 e^{-|3x|} dx$$

The integrand $f(x)$ is continuous on $(-\infty, 0]$.

Integrals with infinite limits of integration are improper integrals of Type I. If $f(x)$ is continuous on $(-\infty, b]$, then $\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$.

To evaluate $\int_{-\infty}^0 e^{-|3x|} dx$, begin by rewriting the integrand based on the relation that for $-\infty < x \leq 0$, $e^{-|x|} = e^x$.

$$\text{So, } \int_{-\infty}^0 e^{-|3x|} dx = \int_{-\infty}^0 e^{3x} dx.$$

Find the antiderivative.

$$\begin{aligned} \int_{-\infty}^0 e^{3x} dx &= \frac{1}{3} \lim_{a \rightarrow -\infty} \int_a^0 3e^{3x} dx \\ &= \frac{1}{3} \lim_{a \rightarrow -\infty} \left[e^{3x} \right]_a^0 \end{aligned}$$

Rewrite e^{3x} as $\frac{1}{3}(3e^{3x})$.

Evaluate the definite integral.

Apply the Fundamental Theorem of Calculus.

$$\begin{aligned} \frac{1}{3} \lim_{a \rightarrow -\infty} \left[e^{3x} \right]_a^0 &= \frac{1}{3} \lim_{a \rightarrow -\infty} (e^{3(0)} - e^{3a}) \\ &= \frac{1}{3} \lim_{a \rightarrow -\infty} (1 - e^{3a}) \end{aligned}$$

Substitute $x = a$ and $x = 0$.

Simplify.

Evaluate the limit.

$$\begin{aligned} \frac{1}{3} \lim_{a \rightarrow -\infty} (1 - e^{3a}) &= \frac{1}{3} \lim_{a \rightarrow -\infty} 1 - \frac{1}{3} \lim_{a \rightarrow -\infty} e^{3a} \\ &= \frac{1}{3}(1) - \frac{1}{3}(0) \\ &= \frac{1}{3} \end{aligned}$$

Evaluate each limit.

Simplify.

Therefore, the value of the original integral is $\frac{1}{3}$.

$$\int_{-\infty}^0 e^{-|3x|} dx = \frac{1}{3}$$